

Smart Metro Ticket Booking System

PROJECT PLANNING PHASE

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

1. Planning Logic

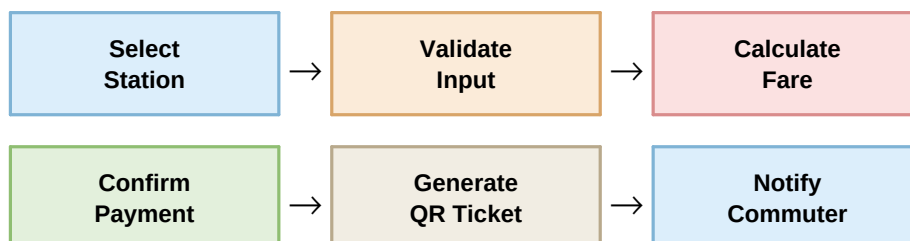
1.1 Overview

The Smart Metro Ticket Booking System is driven by a structured, rule-based logic layer that governs how a commuter's request moves from station selection to digital ticket issuance. Built primarily using ServiceNow Flow Designer, this logic ensures every booking follows a consistent sequence — validating inputs, calculating fares, generating a QR-coded ticket, and notifying the commuter — without manual intervention at any stage.

1.2 Booking Process Logic

- 1. Request Initiation** – Commuter opens the Service Catalog item and enters source station, destination station, travel date, and number of passengers.
- 2. Input Validation** – The system checks that source and destination differ, the travel date is valid, and passenger count is within the allowed range.
- 3. Fare Calculation** – Flow Designer determines the zone/distance between stations and applies the fare rules to compute the total amount.
- 4. Payment Confirmation** – The calculated fare is passed to the payment step for confirmation before the booking is finalized.
- 5. Ticket Generation** – On successful payment, a unique digital ticket with an embedded QR code is generated and linked to the request record.
- 6. Notification Dispatch** – An SMS/Email/Push notification instantly delivers the ticket and booking confirmation to the commuter.
- 7. Record Update** – The Service Catalog request status is updated to “Booked” and stored in the commuter's booking history.

1.3 Logic Flow Diagram



1.4 Validation & Business Rules

- Source and destination stations must not be the same.
- Travel date cannot be earlier than the current date.
- Passenger count must be between 1 and 6 for a single booking request.
- Fare is calculated using zone-based slabs (e.g. Zone 1–2: Rs 10, Zone 3–4: Rs 20, Zone 5+: Rs 30), configurable within Flow Designer.
- A digital ticket remains valid only for the selected travel date and is deactivated automatically after the exit scan or at midnight, whichever comes first.
- Duplicate booking requests for the same station pair, date, and time slot are flagged for review before confirmation.

1.5 Decision Logic by Module

Booking Logic

Captures station pair, travel date, and passenger count; triggers validation before any downstream processing begins.

Fare Logic

Applies zone/distance-based slabs and passenger multipliers to compute an accurate, error-free fare instantly.

Payment Logic

Confirms the transaction against the calculated fare and only allows ticket generation once payment succeeds.

Notification Logic

Selects the delivery channel (SMS/Email/Push) and falls back automatically if the primary channel fails.

1.6 Error Handling & Exception Logic

Scenario	System Response
Invalid station pair selected	Inline error shown; form submission blocked until corrected.
Payment failure	Retry prompt displayed; request record marked "Payment Pending."
Network/API timeout during QR generation	Automatic retry (up to 2 attempts), then escalated to the support queue.
Duplicate booking detected	Request flagged; commuter prompted to confirm before final booking.
Notification delivery failure	System falls back from SMS to Email automatically.

1.7 Logic Summary

The planning logic ensures that every stage of the booking journey — from station selection to ticket delivery — is automated, validated, and consistent. By centralizing fare calculation, validation, and notification rules within Flow Designer, the system removes manual dependency, minimizes fare errors, and guarantees that every commuter receives an accurate, instantly usable digital ticket.